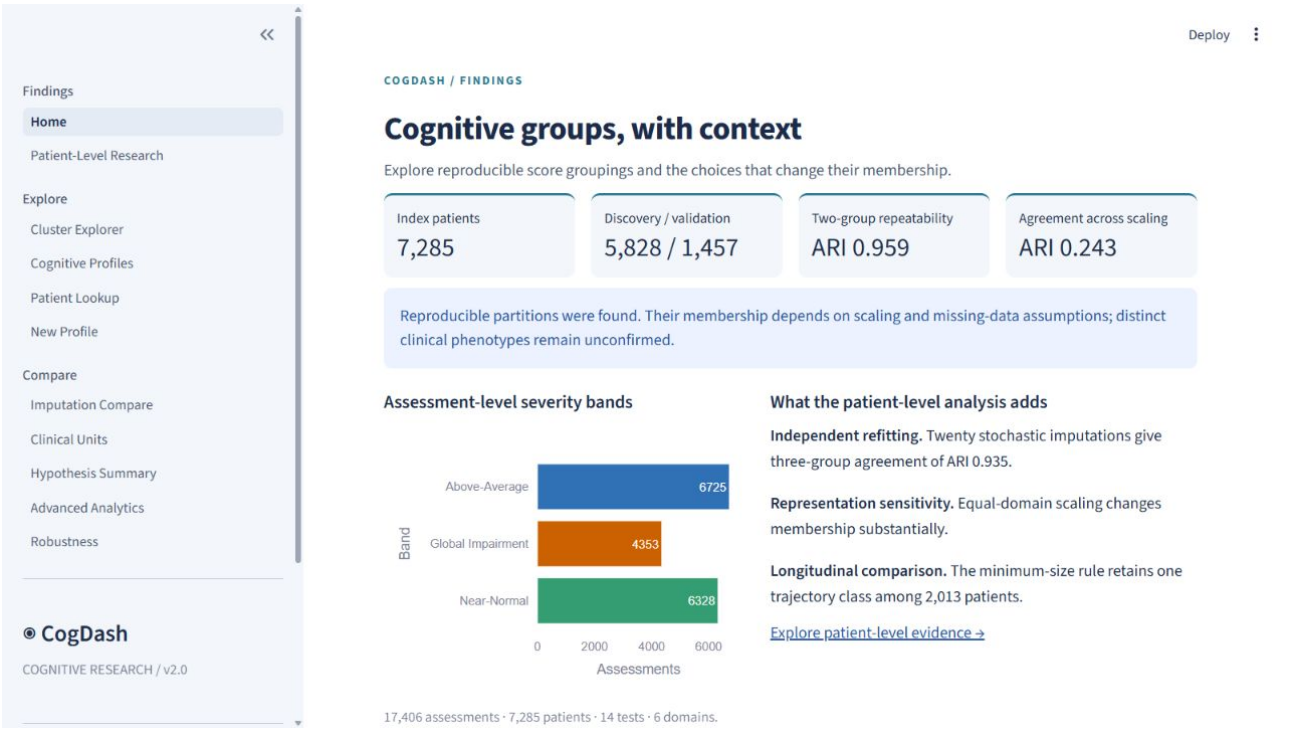
COGNITIVE RESEARCH / VERSION 2.0

CogDash

User Manual

Exploring cognitive score groupings
in acquired brain injury



Reading the evidence. Reproducible partitions were found. Membership depends on scaling and missing-data assumptions; distinct clinical phenotypes remain unconfirmed.

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1 Start here

CogDash presents saved clustering analyses through eleven pages. It reads local results and applies saved transformations for profile comparison. Opening a page does not rerun the experiments.

Launch the prepared workspace

Open PowerShell in the project root and run:

```
./dashboard/run.ps1
```

Open http://127.0.0.1:8501. Keep the terminal running while using CogDash; press Ctrl+C there to stop it. If an instance already occupies that port, reuse it or launch with `-Port 8502`. Fresh-environment setup is at the end of this manual.

Choose a route through the findings

Sidebar area	Pages and purpose
Findings	Home; Patient-Level Research. Main conclusions and the validation design.
Explore	Cluster Explorer; Cognitive Profiles; Patient Lookup; New Profile. Projections, group means and individual profiles.
Compare	Imputation Compare; Clinical Units; Hypothesis Summary; Advanced Analytics; Robustness. Assumptions and supporting evidence.

Use the controls

Choose pages in the sidebar and tabs across the content area. Expand labeled panels for methods and supplementary tables. Scroll the main page to reach lower content; the sidebar and long tables scroll separately.

Hover over plots for values and chart controls. Use zoom, reset and image download where available. Drag to rotate the 3D scatterplot. Tables are readable, scrollable summaries without spreadsheet editing or built-in filtering.

Screenshots are actual browser views captured on 15–16 October 2026. Some show a scrolled detail view. Individual-profile examples are synthetic. The interface may reflow on a smaller screen.

2 Methods and interpretation

Keep the two analysis designs separate

The source extract contains 22,075 assessments from 8,739 patients. Eligibility and variable selection retain 17,406 assessments from 7,285 patients, using 14 tests across six domains. The 50% missingness threshold selects *variables*; an eligible assessment needs at least one observed retained test.

The assessment-level analysis retains repeat visits as separate rows. Its three named bands describe broad cognitive severity within the cohort. Most resampling treats assessments as independent. Clinical Units also reports a separate patient-grouped predictive comparison.

The patient-level analysis uses the earliest eligible assessment per patient. A fixed split assigns 5,828 patients to discovery and 1,457 to validation (seed 20261015). Selection and fitted transformations use discovery data. Validation patients receive those rules without refitting.

Patient-level preprocessing

1. Clip observed tests at discovery 1st and 99th percentiles. Reverse the timed attention test so higher aligned values indicate better performance.
2. Standardize each test using discovery estimates. Apply bounded Bayesian iterative imputation to missing tests.
3. Combine completed tests into six domains using discovery observation-rate weights. Apply robust domain scaling for the main partition; compare equal-domain scaling separately.
4. Fit and select models in discovery data, then evaluate validation assignments and separation. Independent-imputation experiments repeat preprocessing and model fitting for each draw.

Read the principal statistics

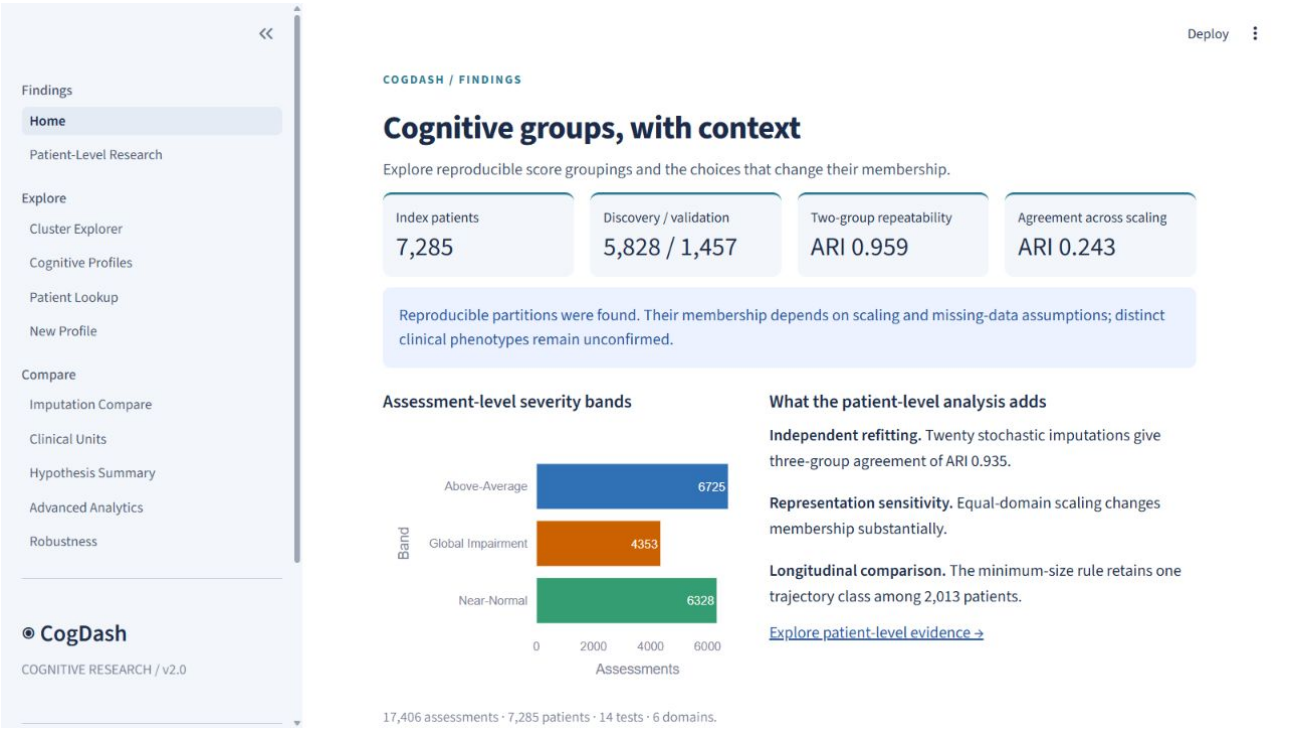
ARI compares partitions after adjusting for chance. One means identical membership; around zero means chance-level agreement. Permuting group labels does not change ARI.

Silhouette measures separation in a specified representation, from -1 to 1. Higher values indicate better separation under that geometry, without establishing a clinical category.

BIC balances fit and complexity; lower is preferred for comparable models. Higher validation log likelihood indicates better held-out density fit on a common measurement basis.

Group numbers and colors belong to a particular solution. Matching numbers across models do not identify the same people. Cohort z-scores have no external normative calibration.

3 Home



Home combines patient-level findings with historical assessment-level bands.

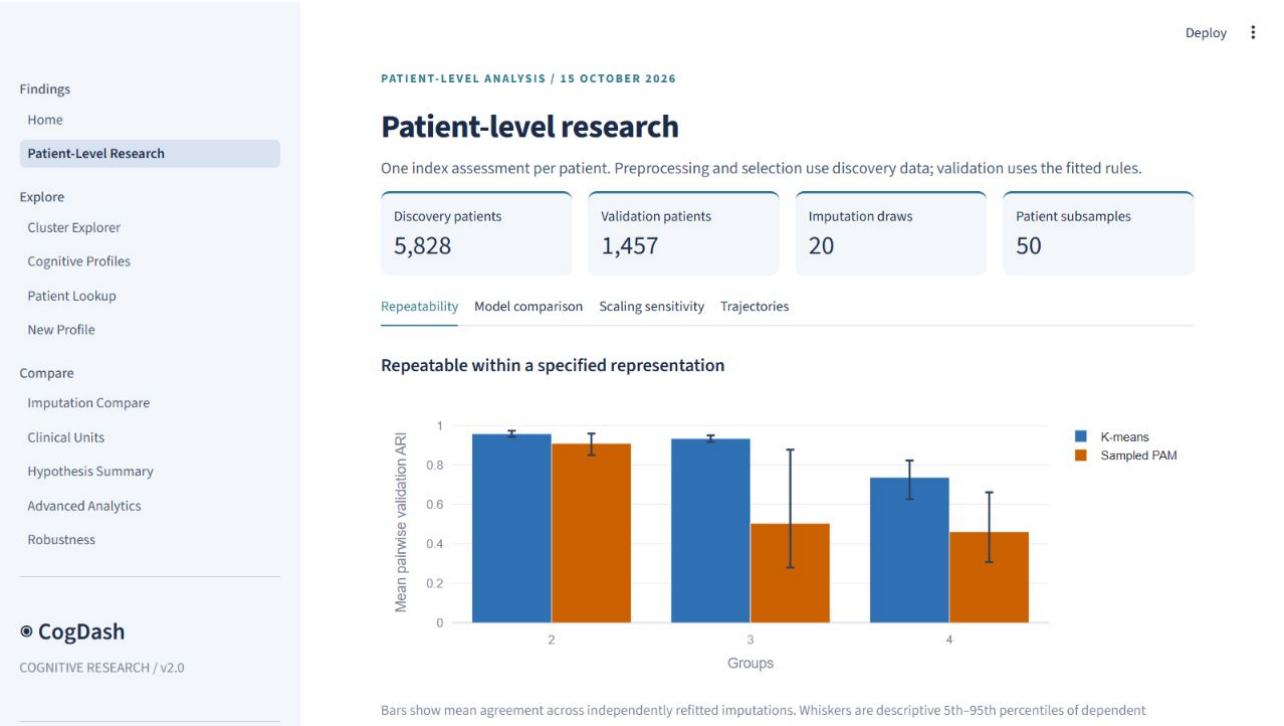
Read the four cards first: 7,285 index patients, the discovery/validation split, two-group repeatability of ARI 0.959, and agreement across scaling of ARI 0.243.

The assessment-level chart shows the original named MICE groups: 6,725 Above Average, 4,353 Globally Impaired and 6,328 Near Normal assessments. These names describe the cohort partition without diagnostic or normative validation.

Select **Explore patient-level evidence** or **Patient-Level Research** to inspect the new analyses. Expand **Hypothesis interpretation** for the H1–H4 summary.

> Reproducible score partitions were found under specified modeling choices. The current evidence does not establish distinct clinical phenotypes.

4 Patient-Level Research: repeatability



Repeatability tab. Bars summarize independent stochastic-imputation fits.

Choose **Patient-Level Research**, then **Repeatability**. The chart compares pairwise validation assignments across twenty independently refitted imputations for two- and three-group solutions.

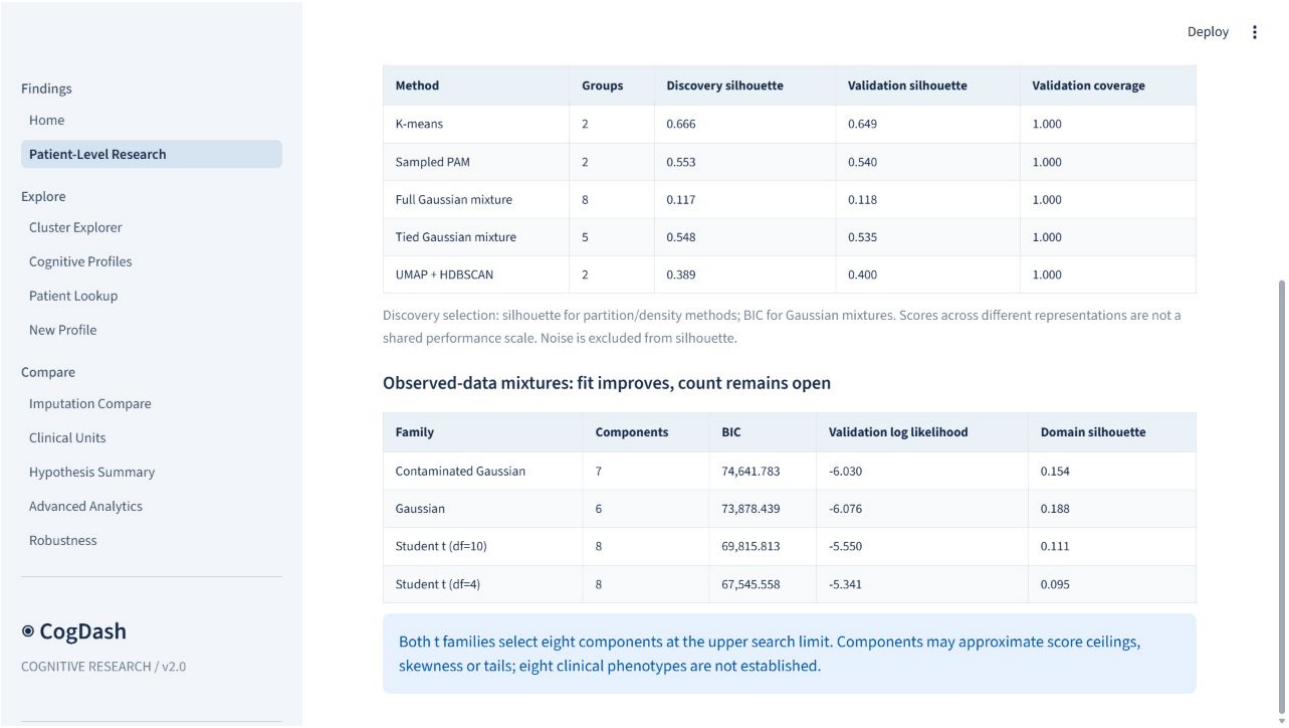
K-means mean agreement is ARI 0.959 with two groups and 0.935 with three. Two groups were selected in all twenty silhouette screens over two to eight groups. This search does not test a one-population alternative.

Whiskers show the 5th–95th percentiles of pairwise agreement. Pairs share fitted partitions, so these are descriptive ranges, not confidence intervals.

Expand **Methods and additional stability results** for the full table. Three-group k-means agreement is approximately 0.968 across fifty patient subsamples and 1.000 across initialization checks.

Repeatability is conditional on the representation and missing-data model. Compare Scaling sensitivity before interpreting the groups.

5 Patient-Level Research: model comparison



Scrolled detail with selected robust-domain models and observed-data mixtures.

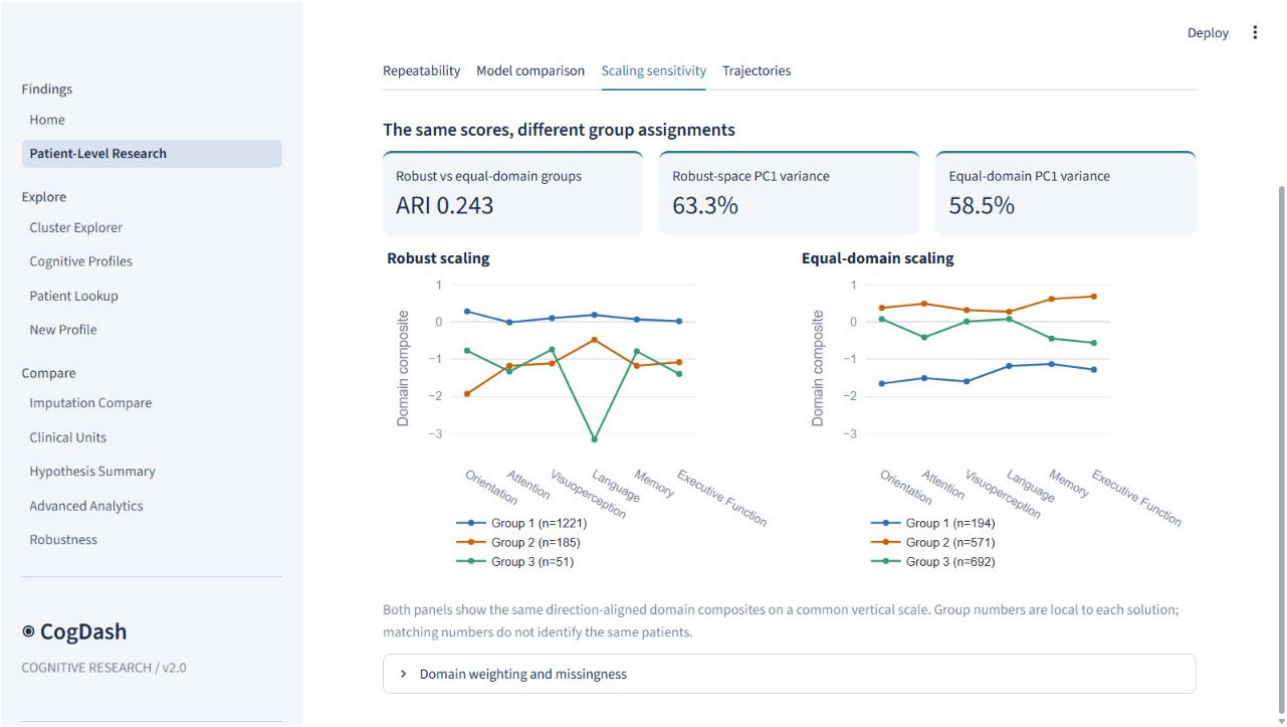
Choose **Model comparison**. Use **Representation** to switch between robust domains, PC1 residuals, fourteen tests and the representation without orientation. Clear **Show discovery-selected models only** to inspect candidate fits.

Two-group k-means in robust domains has discovery/validation silhouettes of 0.666/0.649. Coverage is the fraction of validation observations assigned; silhouette excludes noise observations.

The lower table compares mixtures fitted to observed entries. The Student t family with four degrees of freedom has BIC 67,545.6 and validation log likelihood -5.341. Both Student t families select eight components, the upper search limit.

Components can approximate skewness, tails or score ceilings. Eight selected components do not establish eight clinical phenotypes. Read every score with its measurement space and selection rule.

6 Patient-Level Research: scaling sensitivity



Scaling sensitivity detail. Both panels use common domain-composite axes.

Select **Scaling sensitivity**. The robust and equal-domain three-group profiles show validation group means, with sizes in the legends. Both panels use the same vertical scale and direction-aligned composites.

Membership agreement is only ARI 0.243. Robust scaling produces groups of 1,221, 185 and 51 validation patients; equal-domain scaling produces 194, 571 and 692. This is a substantial partition change using the same underlying scores.

The leading principal component explains 63.3% of robust-space variance and 58.5% with equal-domain scaling. Expand **Domain weighting and missingness** for weights and missingness associations.

Group 1 in one panel is not automatically Group 1 in the other. Missingness associations do not establish causation, but show why repeatability under one imputation model is insufficient evidence of biological discreteness.

7 Patient-Level Research: trajectories



Scrolled trajectory detail with class sizes, fit criteria and the selection rule.

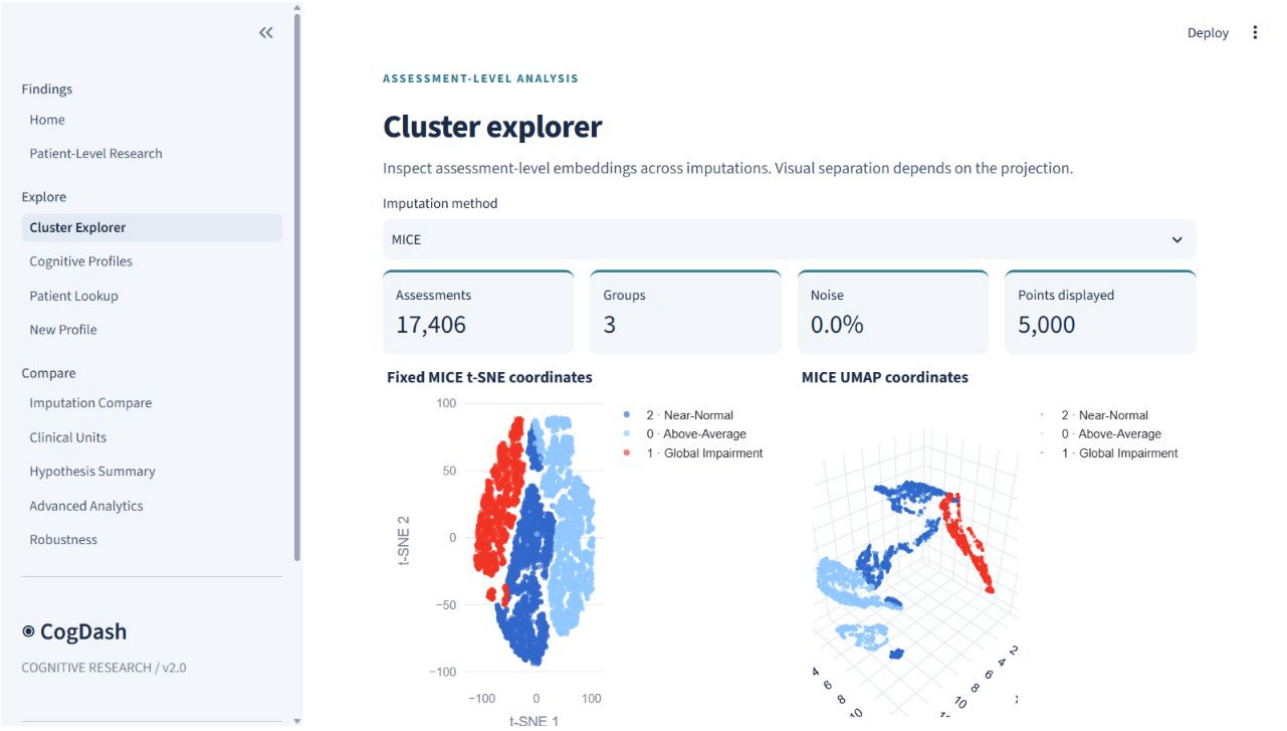
Select **Trajectories**. The analysis includes 2,013 patients with at least three sufficiently measured visits across 0.25–5 years: 1,590 discovery and 423 validation patients.

Linear random-intercept/slope mixtures compare one to four classes. The dashed line marks the 5% minimum expected class-size rule. Although fit improves with more classes, the smallest discovery classes fall to 3.9%, 2.7% and 1.3% for the two-, three- and four-class fits.

The rule therefore retains one class. The table gives BIC, validation log likelihood and the smallest expected class size for each candidate.

One selected class supports continuous variation under this design. It does not exclude small subgroups, nonlinear change, or different results under another model specification.

8 Cluster Explorer



MICE labels in fixed t-SNE coordinates and the method-specific 3D UMAP projection.

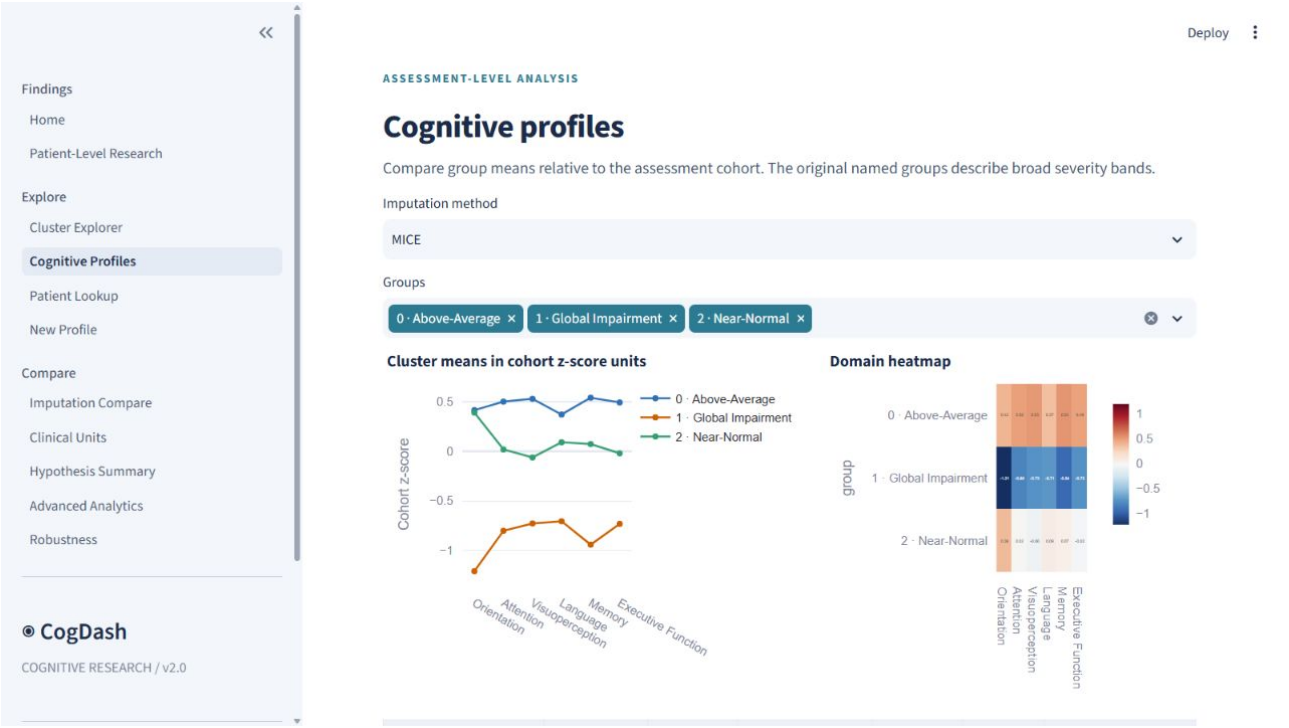
Choose **Cluster Explorer**, then an **Imputation method**. Cards report assessment count, recovered groups, noise fraction and displayed points.

The left plot uses fixed MICE t-SNE coordinates, recolored with the selected partition. The right plot uses that method's UMAP coordinates. Hover over points, zoom the 2D view, or drag to rotate the 3D view. Use the toolbar to reset.

A deterministic sample of up to 5,000 rows keeps display responsive. Counts refer to the full data. Every point is an assessment; repeat visits remain separate.

Changing imputation changes labels and UMAP coordinates. Visual separation in an embedding does not by itself demonstrate separation in the measured cognitive domains.

9 Cognitive Profiles



MICE group means as domain profiles and a cohort-standardized heatmap.

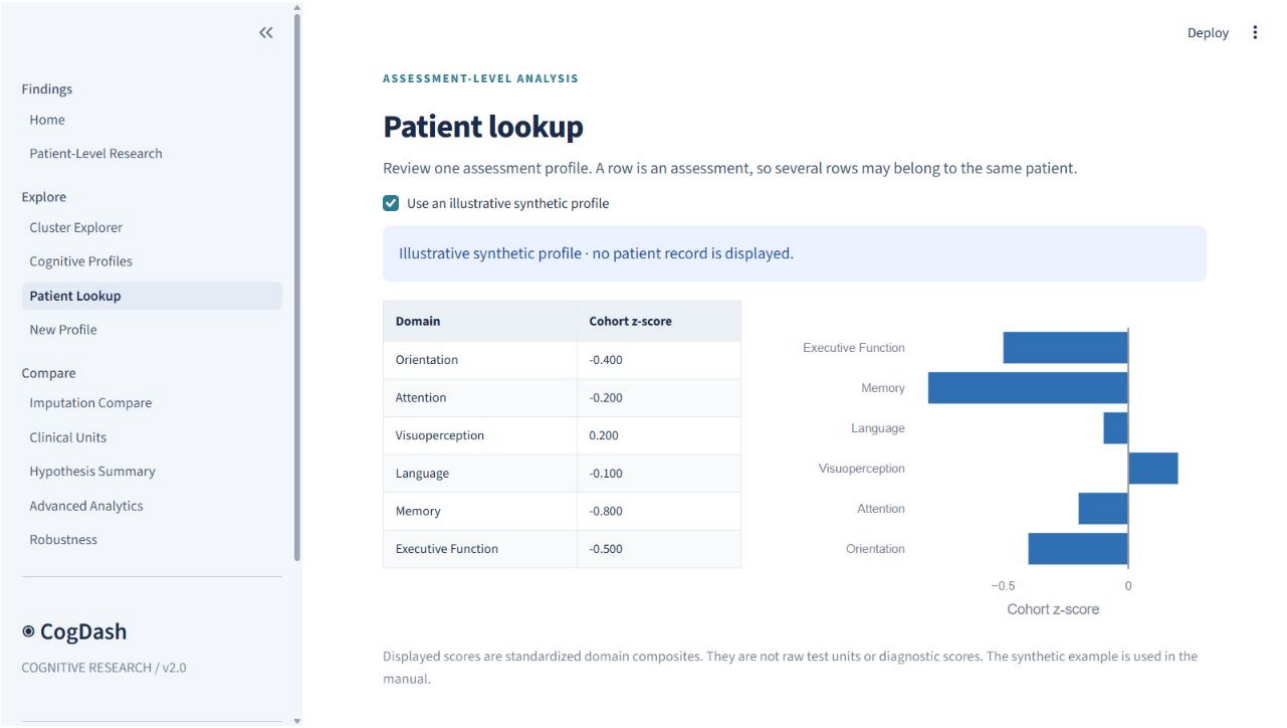
Choose an **Imputation method**, then add or remove entries in **Groups**. Clearing every group displays a prompt to select at least one.

The line chart compares six mean domain profiles. The heatmap shows the same values numerically and by color. Scroll down for the table. Zero is the assessment cohort mean after standardization; positive and negative values indicate position within this cohort.

Use profile shape to identify domains that differ between groups. Read magnitude alongside the severity gradient and preprocessing design. Labels and colors apply to the selected model.

These are cohort z-scores, not external age-, education- or population-adjusted norms. Historical group names are descriptive, without diagnostic cutoffs.

10 Patient Lookup



Default synthetic profile. No patient record appears in this screenshot.

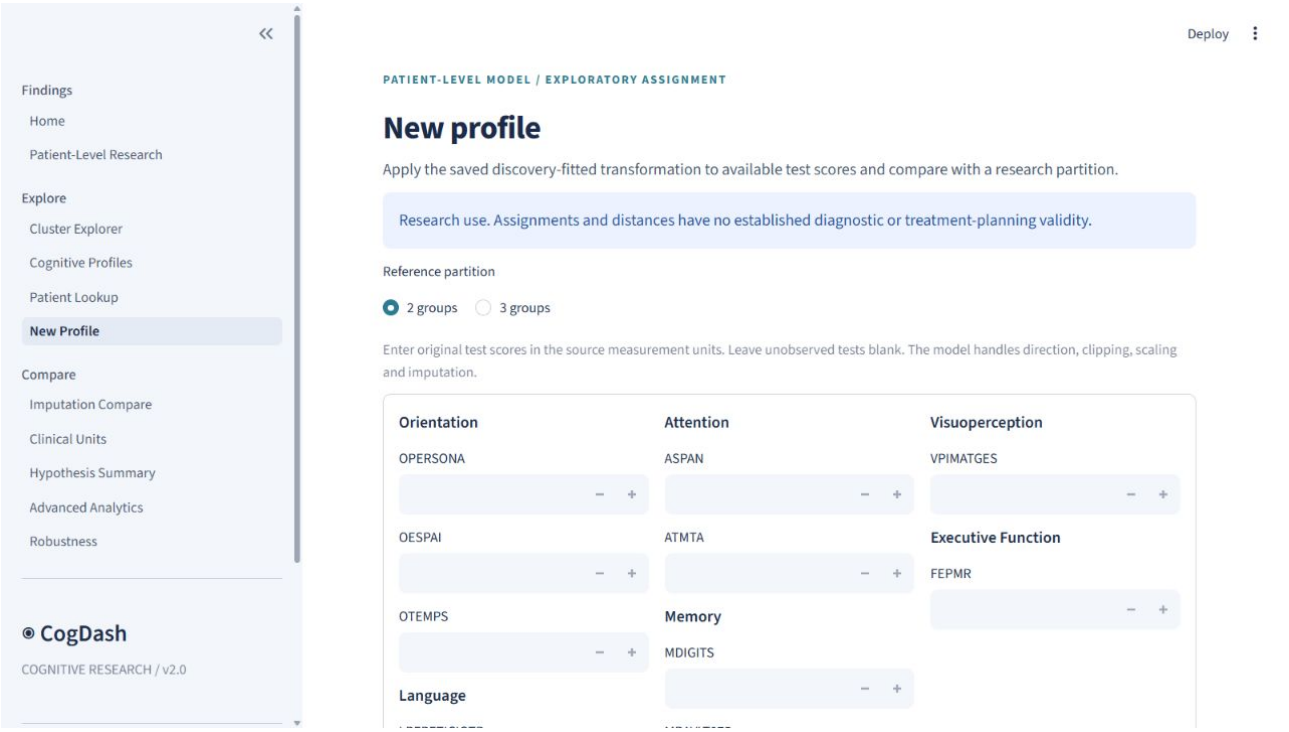
The default **Use an illustrative synthetic profile** checkbox displays six example domain values and a bar chart. The information panel identifies the example as synthetic.

For a local assessment review, clear the checkbox and enter an **Assessment row**, from zero through 17,405. The page displays its saved MICE band and cohort-standardized domain profile.

A row number is an assessment position, not a patient identifier. Multiple assessments may belong to one patient. This page retrieves existing assessment-level results.

Displayed values are standardized domain composites. Use New Profile to enter original test scores for exploratory comparison with a saved patient-level partition.

11 New Profile: enter scores



Upper form. Scroll to reach the remaining tests and Compare profile.

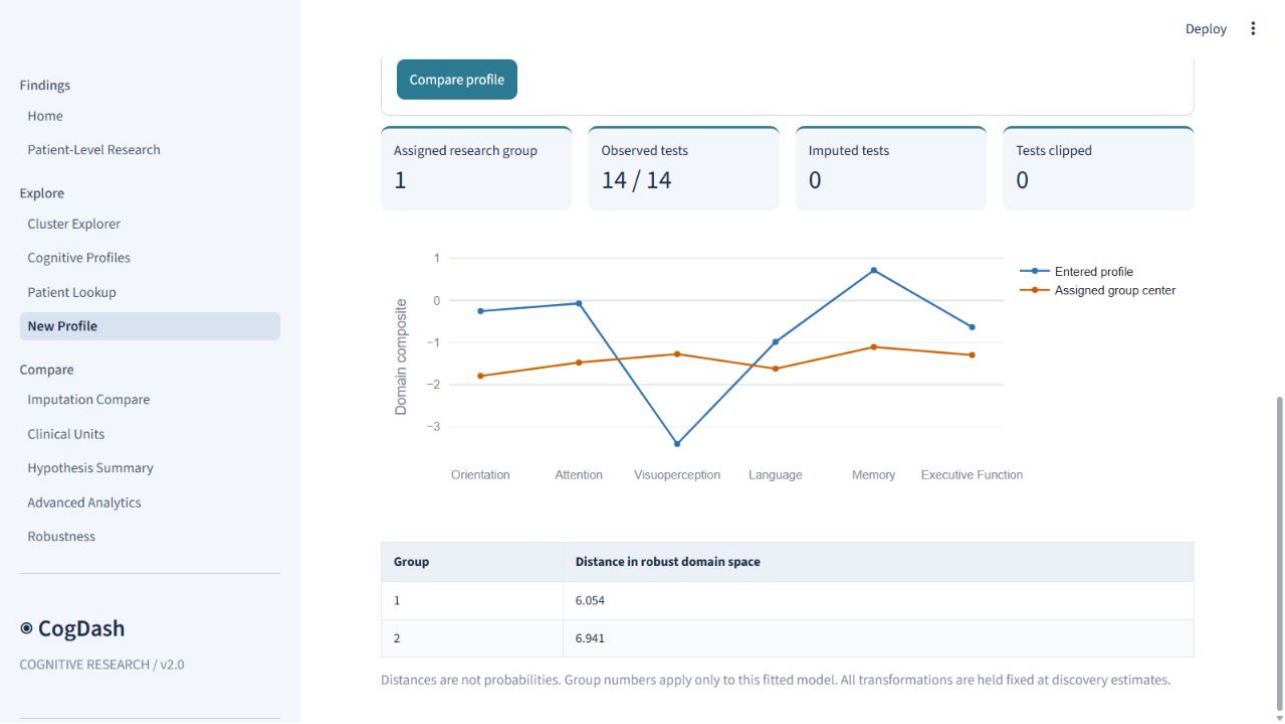
Select **New Profile**, then the **Reference partition**: two or three groups. Enter available scores in original measurement units. Fourteen fields cover orientation, attention, visuoperception, language, memory and executive function.

Leave unobserved tests blank. A blank is missing; zero is an observed score. Do not pre-standardize, reverse the timed attention score, or calculate domain averages yourself.

Scroll down and select **Compare profile**. Saved discovery transformations handle clipping, direction, test scaling, bounded imputation, domain aggregation and robust domain scaling. Entered profiles do not refit the model.

An empty form is rejected. Fewer than seven observed tests triggers an imputation-dependence warning. Scores outside saved discovery bounds are clipped and the affected tests are reported.

12 New Profile: read the result



Synthetic complete profile using the two-group reference model.

Cards show the assigned research group and observed, imputed and clipped test counts. The chart compares transformed input with the assigned center in direction-aligned domain-composite units. The table gives distances to centers in robust domain space.

The synthetic example produces Group 1, with fourteen observed tests, no imputation and no clipping. Distances are approximately 6.054 and 6.941. Lower distance indicates the nearer center; it is not a probability or confidence score.

Reproduce the example. Enter orientation scores 6, 5 and 20; ASPAN 5; ATMTA 60; VPIMATGES 8; each language score 10; MDIGITS 6; MRAVLT075 35; MRAVLT015 6; MRAVLT015R 12; and FEPMR 18.

This is exploratory comparison with a saved research partition. Group membership and distance have no established diagnostic or treatment-planning validity.

13 Imputation Compare



Pairwise ARI and NMI for the fixed-reference assignment comparison.

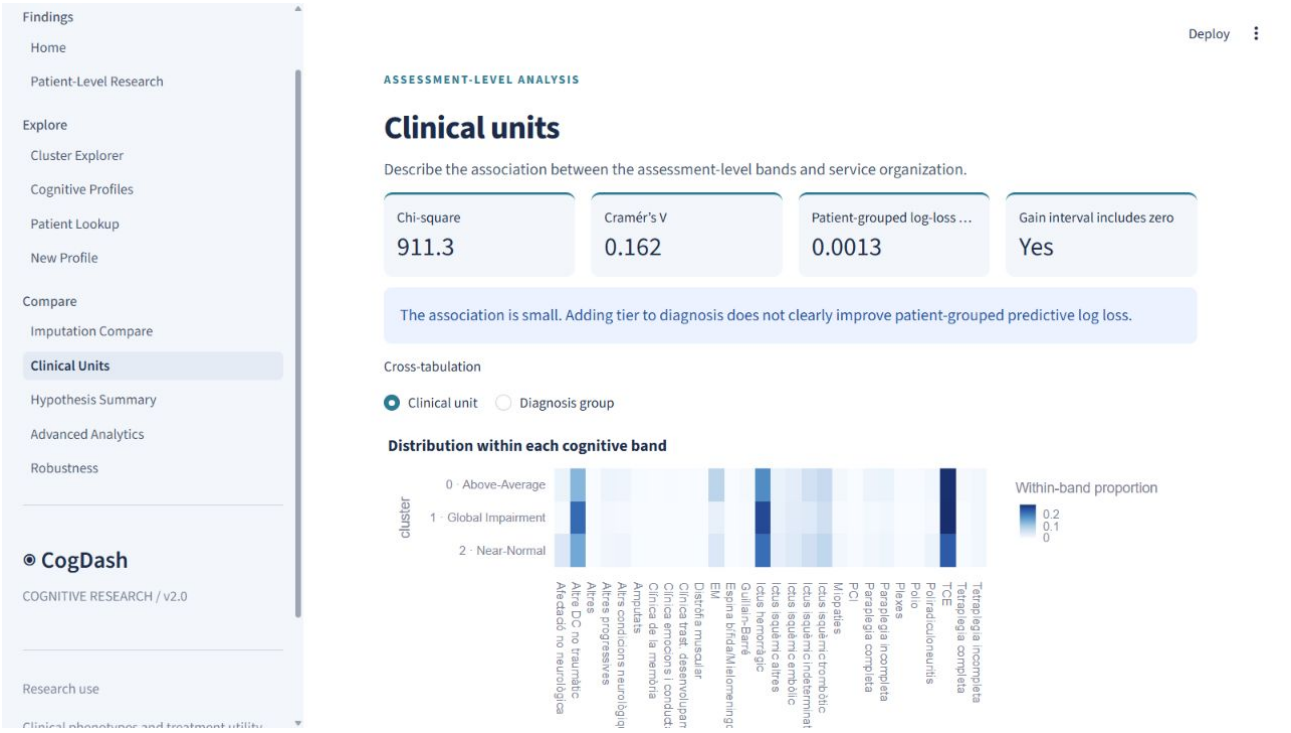
Read the ARI matrix on the left and normalized mutual information (NMI) on the right. Higher values indicate greater agreement.

Cards summarize mean ARI 0.710, median ARI 0.739, consensus agreement 0.929, and 86.2% of assessments with at least 80% agreement. Expand **Method metrics and domain divergence** for supplementary tables.

The same MICE-fitted transformation and three-group partition are applied across imputations. This evaluates assignment sensitivity with the reference structure held fixed.

Consensus agreement is not a calibrated clinical probability. Patient-Level Research reports independently refitted preprocessing and clustering pipelines.

14 Clinical Units



Clinical-unit distributions within each assessment-level cognitive band.

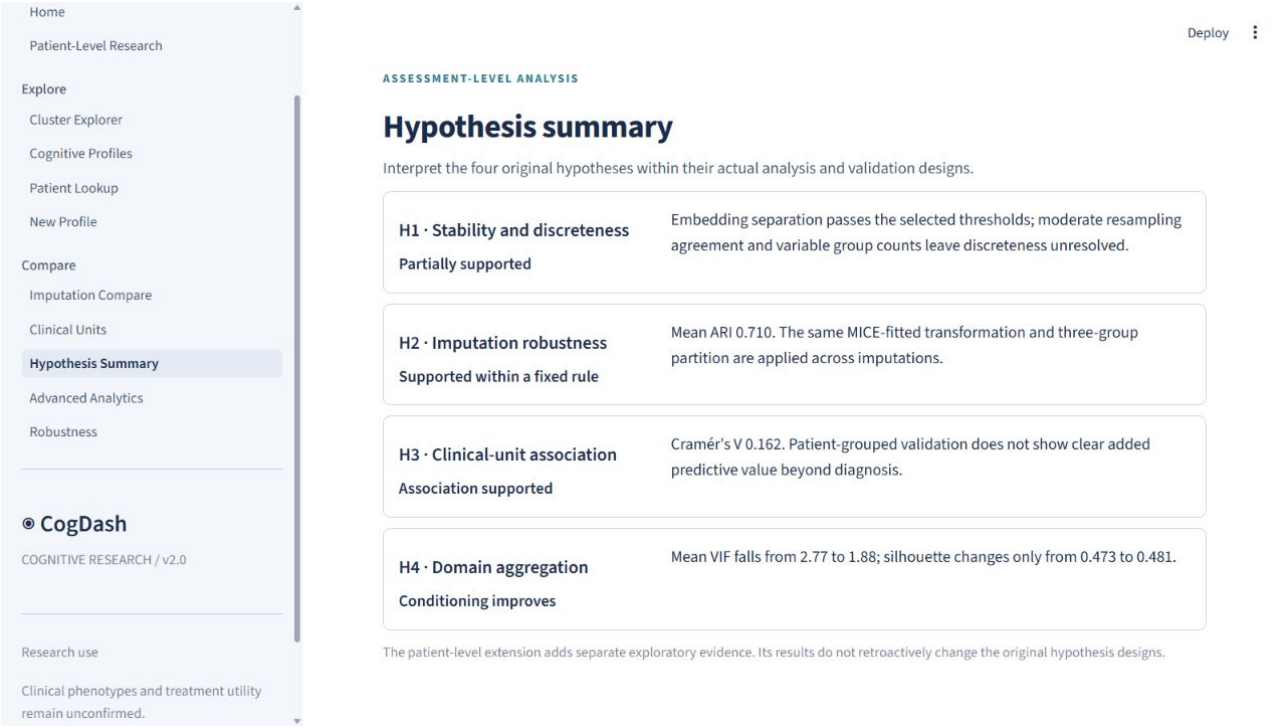
Switch **Cross-tabulation** between **Clinical unit** and **Diagnosis group**. The heatmap shows proportions within each band. Expand **Counts and grouped-validation results** for counts and the predictive comparison.

The assessment-level association has chi-square 911.3 and Cramér's $V = 0.162$: statistically detectable but small. The patient-grouped comparison asks whether cognitive tier adds predictive information beyond diagnosis.

Log-loss improvement is approximately 0.0013, with a 95% interval from -0.00064 to 0.00307. Clear added predictive value is not established because the interval includes zero.

Association with service organization does not establish appropriate placement, treatment benefit, or absence of other confounding.

15 Hypothesis Summary



Hypothesis cards with conclusions matched to their analysis designs.

H1: Stability and discreteness. Partially supported. Embedding thresholds are met; moderate resampling agreement and variable group counts leave discreteness unresolved.

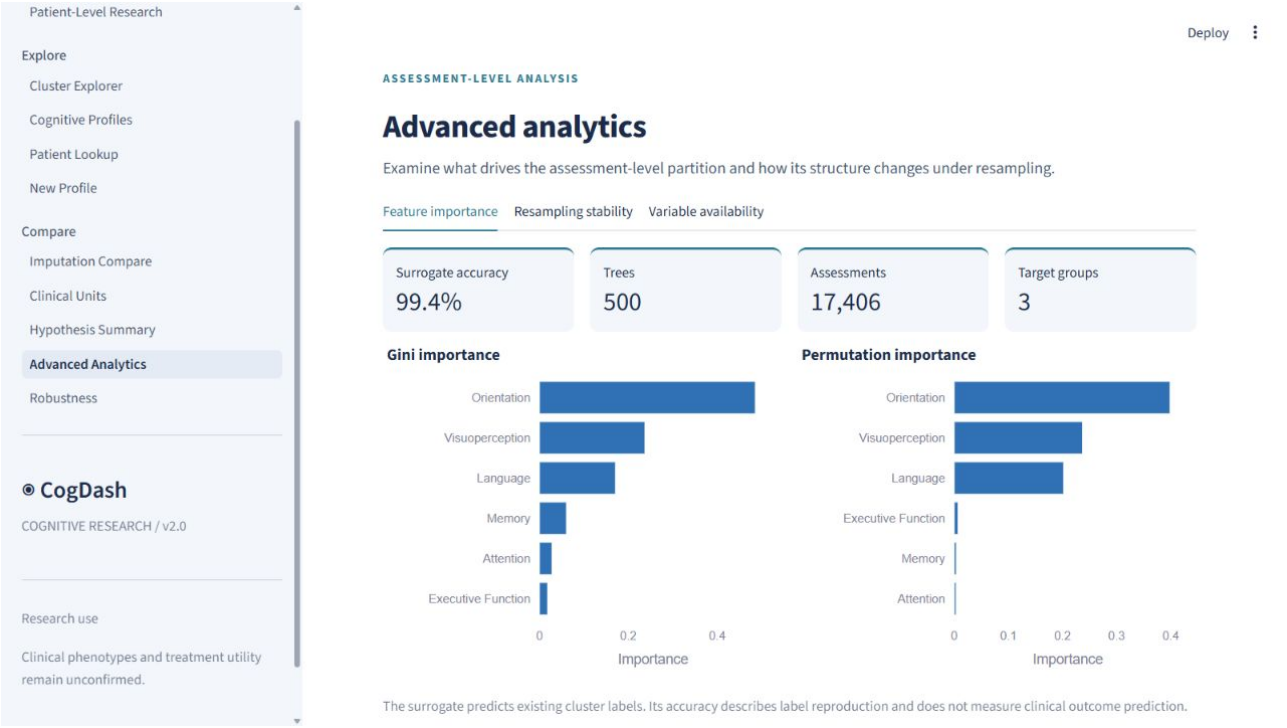
H2: Imputation robustness. Supported within a fixed rule. Mean ARI is 0.710 with the same MICE-fitted structure across imputations.

H3: Clinical-unit association. Association supported. The effect is small; patient-grouped validation shows no clear added predictive value beyond diagnosis.

H4: Domain aggregation. Conditioning improves. Mean VIF falls from 2.77 to 1.88; silhouette changes only from 0.473 to 0.481.

The patient-level extension supplies separate exploratory evidence. It does not retroactively change the original hypothesis designs.

16 Advanced Analytics



Feature importance for a surrogate of existing assessment-level labels.

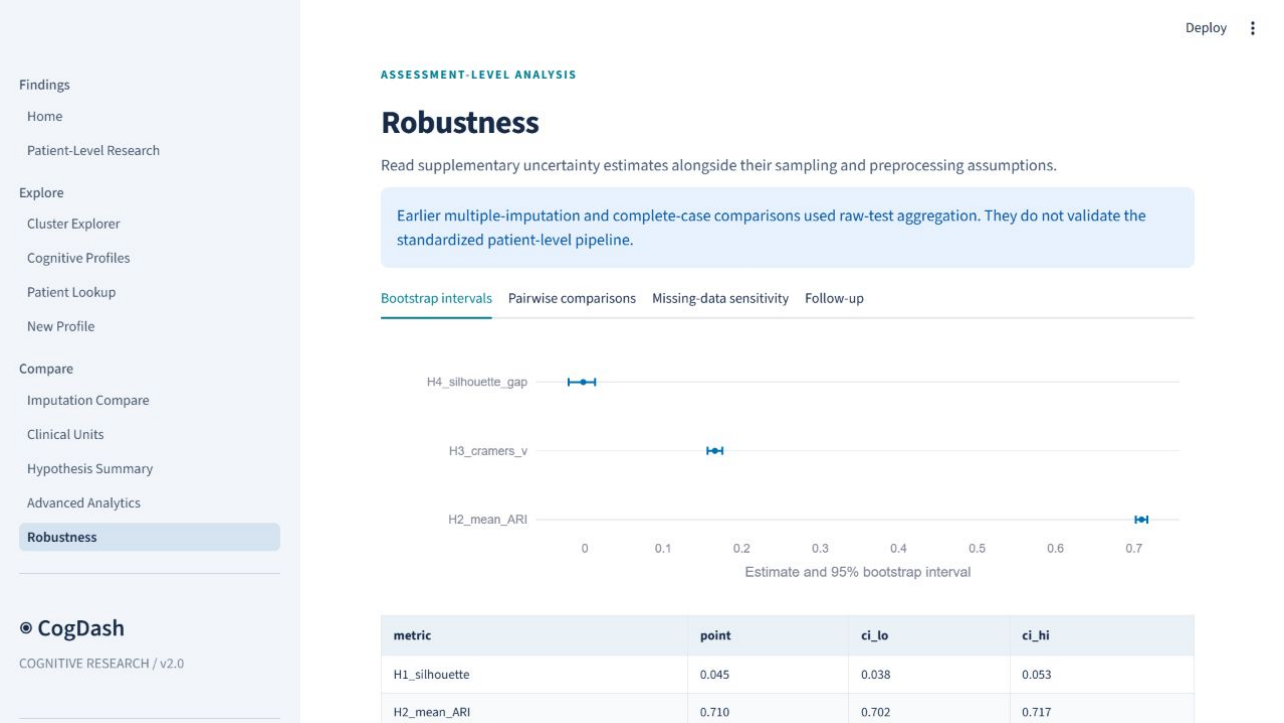
Feature importance compares Gini and permutation importance. A 500-tree random forest reproduces the three labels with approximately 99.4% cross-validated accuracy. This is label reproduction, not medical outcome prediction. Orientation contributes strongly.

Resampling stability displays pairwise ARI and summary statistics. It samples assessment rows without grouping repeat visits. Patient-level resampling is reported separately.

Variable availability plots silhouette against maximum variable missingness and lists candidate results. The threshold selects tests, not rows. A silhouette of -1 accompanies the no-cluster solution.

Importance identifies inputs that reproduce this labeling. It does not establish causes of impairment or the clinical importance of a test.

17 Robustness



Bootstrap intervals tab; scroll for the complete table and sampling qualification.

Bootstrap intervals resample assessments and ignore within-patient dependence. Supplementary estimates can differ from headline values because of their own sampling and preprocessing basis.

Pairwise comparisons presents multiplicity-adjusted tests within the fixed-reference design. Repeated pairwise tests do not establish discrete phenotypes.

Missing-data sensitivity shows descriptive summaries across five MICE draws and a complete-assessment comparison. Within-imputation variances were not estimated; intervals are not Rubin-rule confidence intervals.

Follow-up shows time-since-injury and within-patient transitions. These lack an independent functional endpoint and do not establish causal recovery or treatment benefit.

Earlier multiple-imputation and complete-case comparisons used raw-test aggregation. They do not validate the standardized patient-level pipeline.

18 Installation, data and troubleshooting

Fresh environment

Use Python 3.12. From the project root, create the environment:

```
py -3.12 -m venv .venv-research
```

Use the environment's explicit Python path to install requirements and prepare assets:

```
$py = ".\.venv-research\Scripts\python.exe"
& $py -m pip install -r analysis/requirements-research.lock.txt
& $py -m pip install -r dashboard/requirements.txt
& $py -m dashboard.export_data
```

Launch with `./dashboard/run.ps1`. The current workspace is already prepared. The setup guide is in `dashboard/README.md`.

Data contract

CogDash needs saved assessment-level outputs in `results/` and patient-level outputs in `results/research_2026/`. These generated directories are excluded from Git; a source checkout alone is insufficient.

Asset preparation reads saved models and eligible data, verifies assignments, and writes a compact inference model, aggregate clinical-unit tables, validation profiles and source hashes into `results/dashboard/`. It does not train models. Use trusted local model files. Regenerate assets when the analysis changes, then restart CogDash.

Common issues

Symptom	Action
Missing-results message	Restore saved results and run asset preparation. The setup guide lists required inputs.
Port already occupied	Reuse the instance or launch with `-Port 8502`.
Old values remain visible	Restart after updating results or application code.
3D plot unavailable	Use a browser with WebGL support. The 2D view and profile tables remain available.
Empty profile rejected	Enter at least one observed raw score. Leave missing tests blank.

Verification and source

Seven functional tests cover all eleven pages, data consistency, model assignment, input validation and key controls. Using the Python path set above, run:

```
& $py -m unittest dashboard.tests.test_dashboard -v
```

Source is in `dashboard/`; screenshots and their capture manifest accompany this manual.